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Deng et al.

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(54) **SEMICONDUCTOR STRUCTURE AND MANUFACTURING METHOD THEREOF**

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(58) **Field of Classification Search**

None

See application file for complete search history.

(56)

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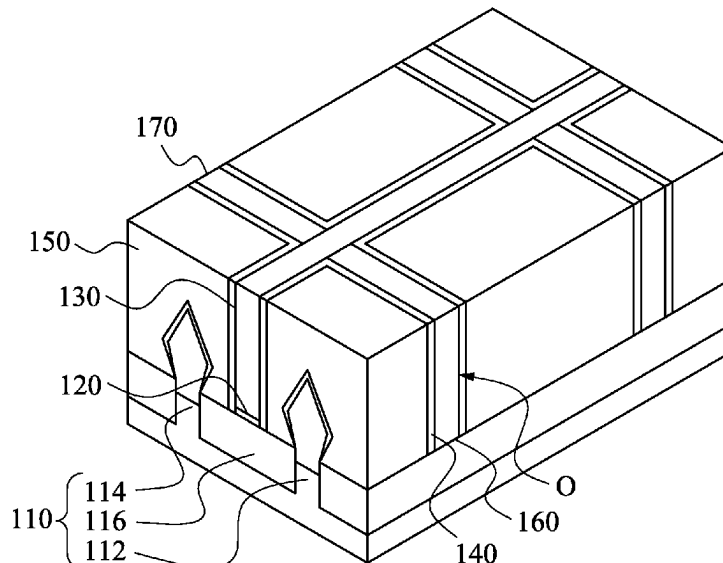
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(57)

ABSTRACT

A semiconductor structure includes a first fin, a second fin, a first gate, a second gate, at least one spacer, and an insulating structure. The first gate is present on the first fin. The second gate is present on the second fin. The spacer is present on at least one side wall of at least one of the first gate and the second gate. The insulating structure is present between the first fin and the second fin, in which the spacer is substantially absent between the insulating structure and said at least one of the first gate and the second gate.

20 Claims, 12 Drawing Sheets



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division of application No. 15/061,876, filed on Mar.
4, 2016, now Pat. No. 10,510,608.

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H10D 89/10 (2025.01)

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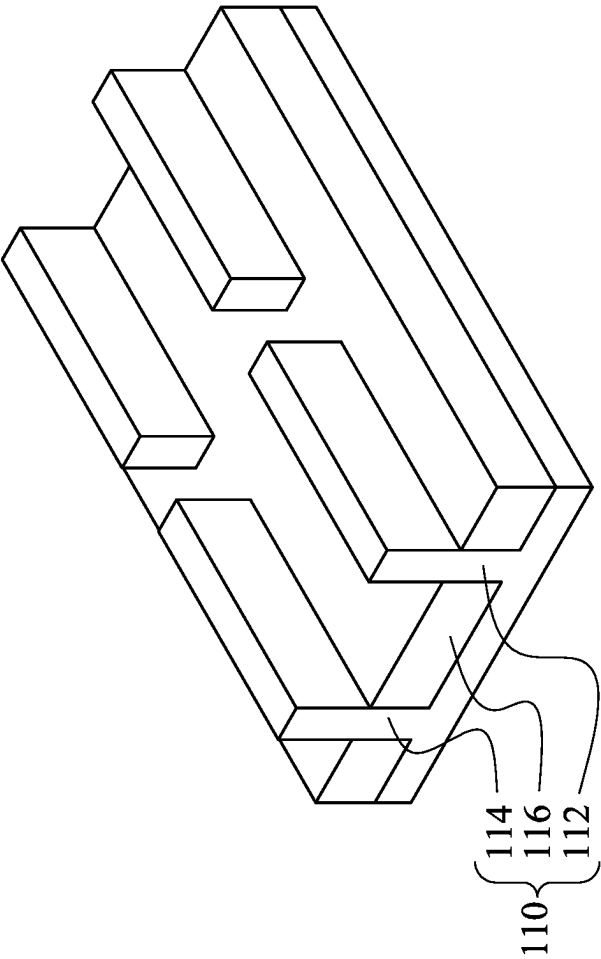


Fig. 1

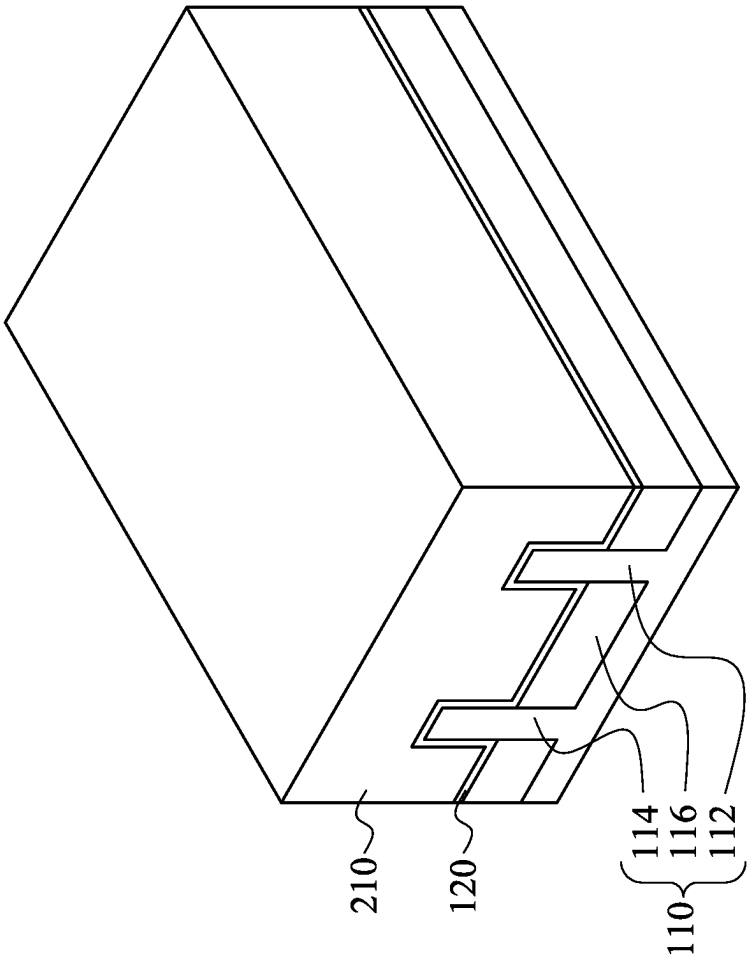


Fig. 2

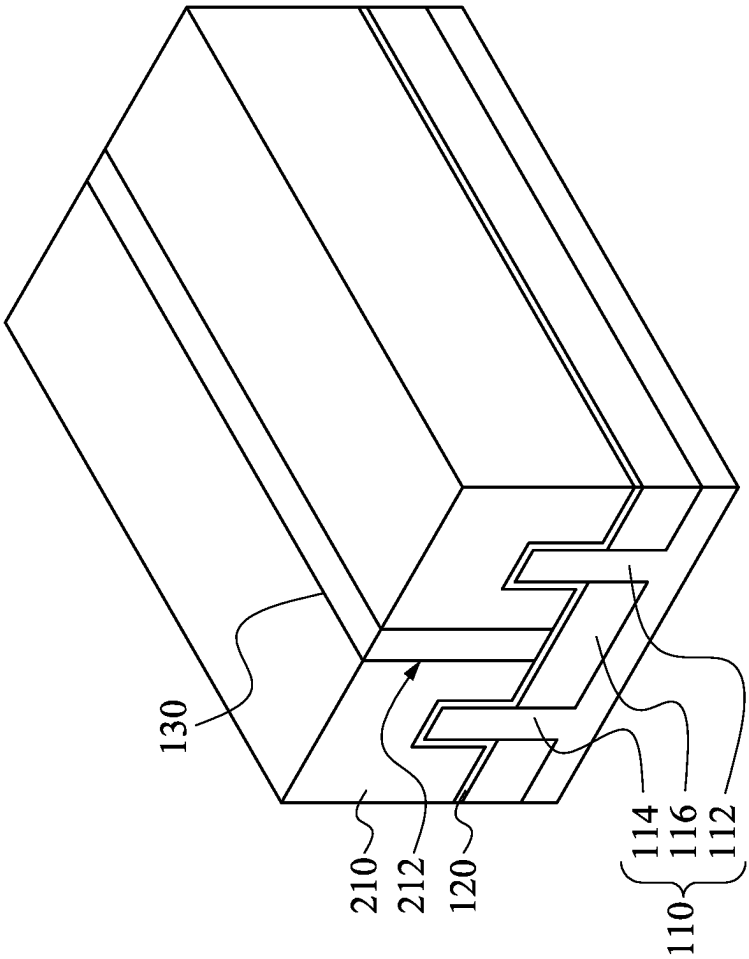


Fig. 3

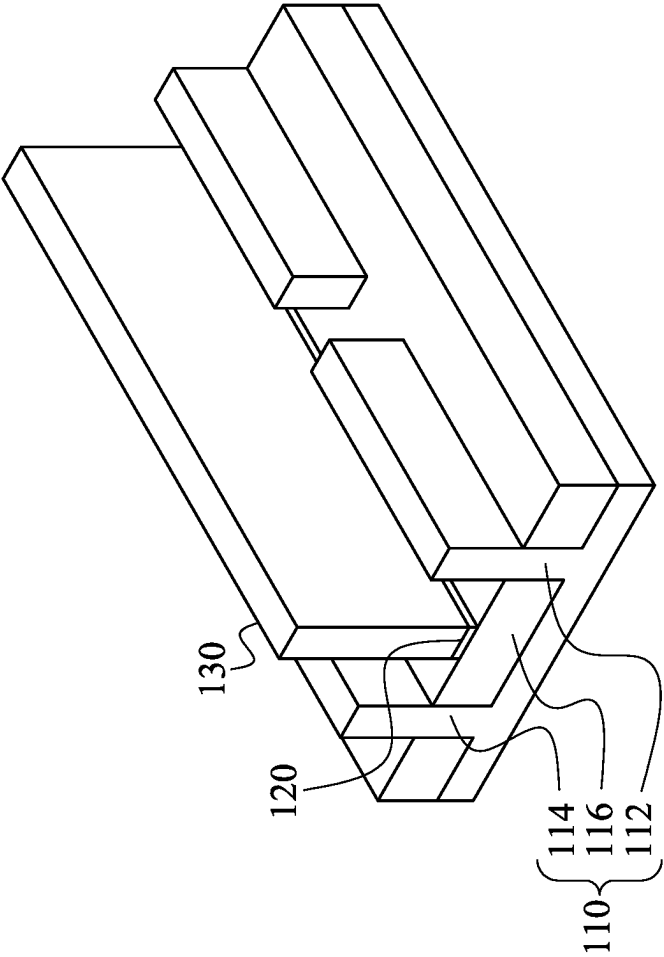


Fig. 4

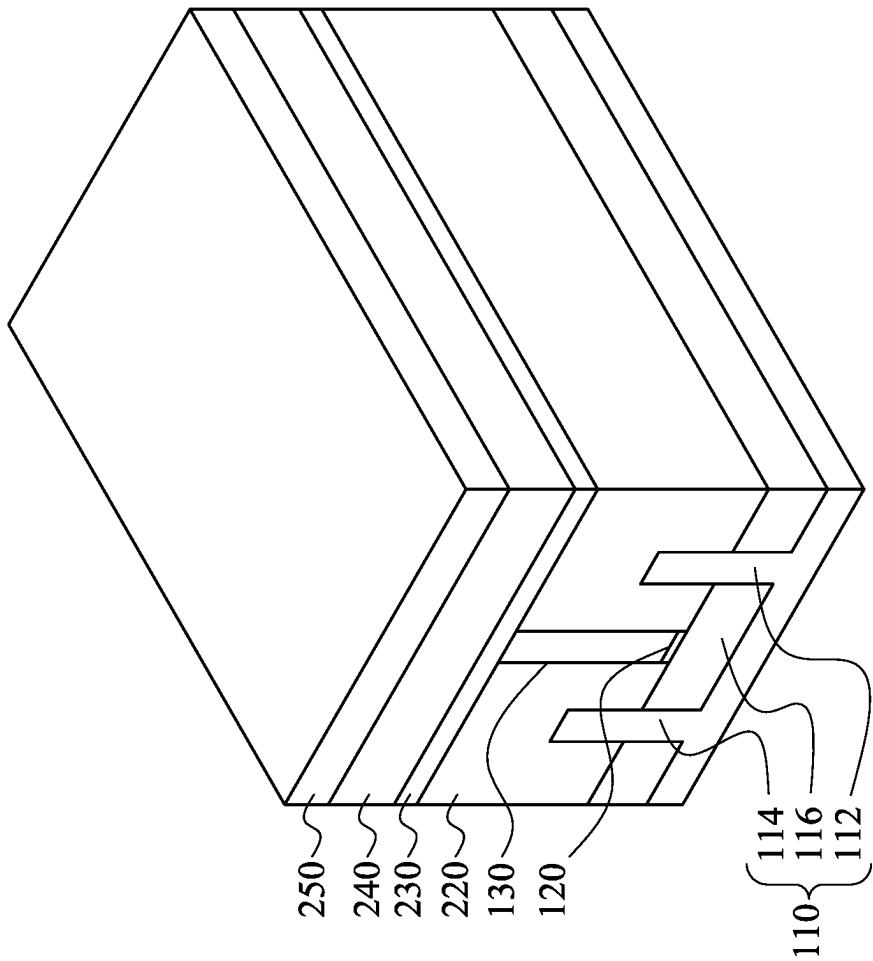


Fig. 5

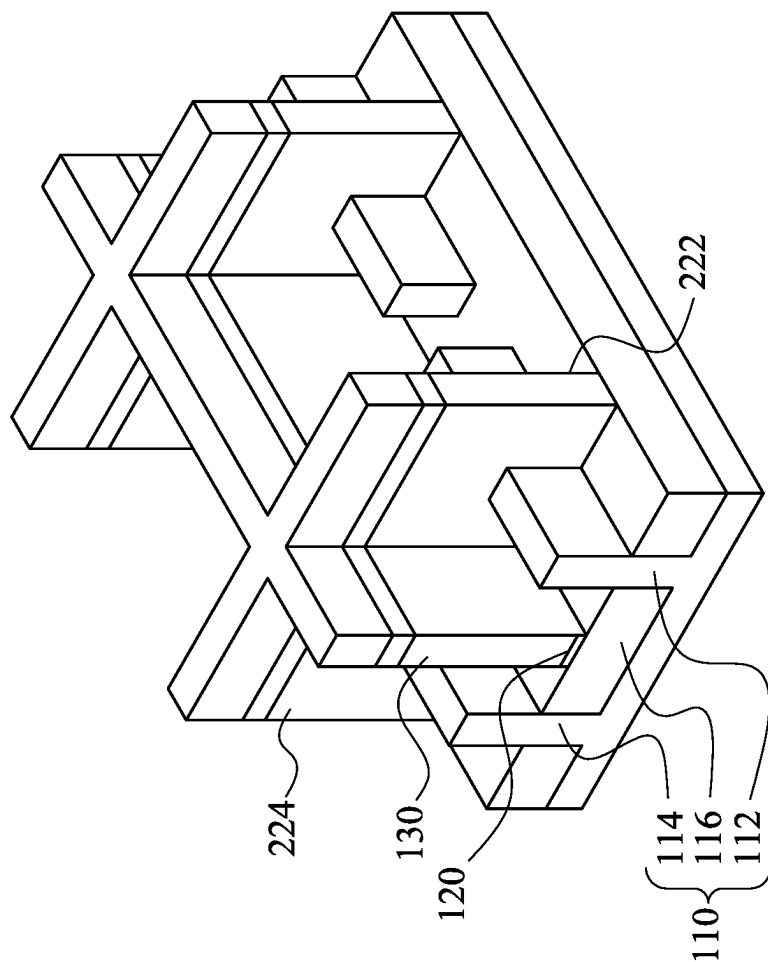


Fig. 6

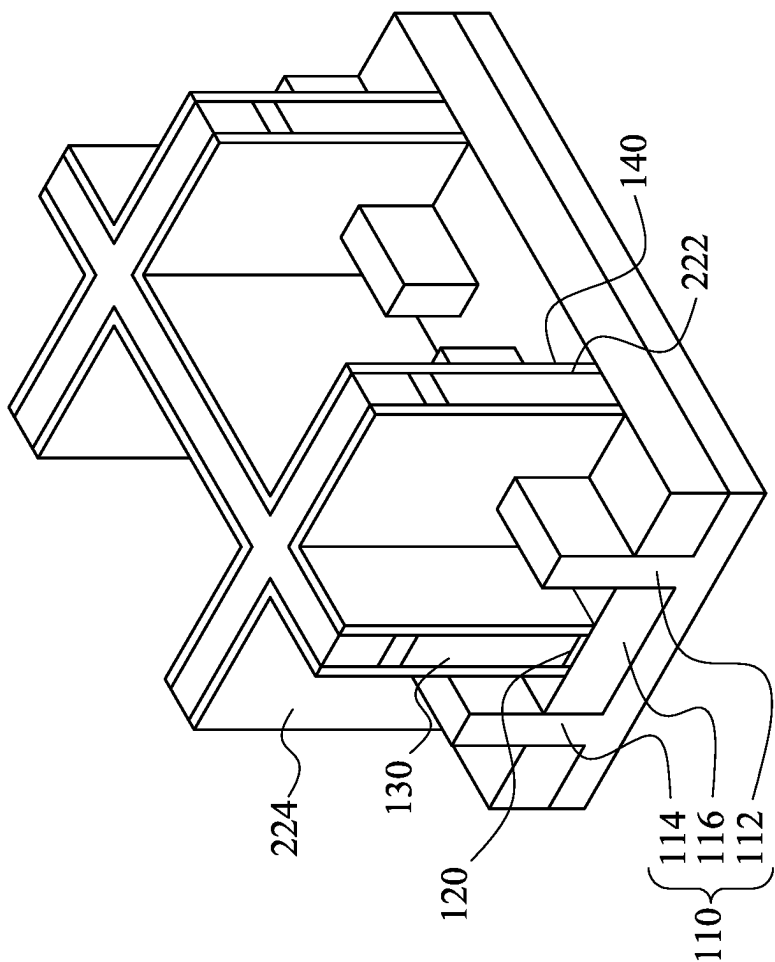


Fig. 7

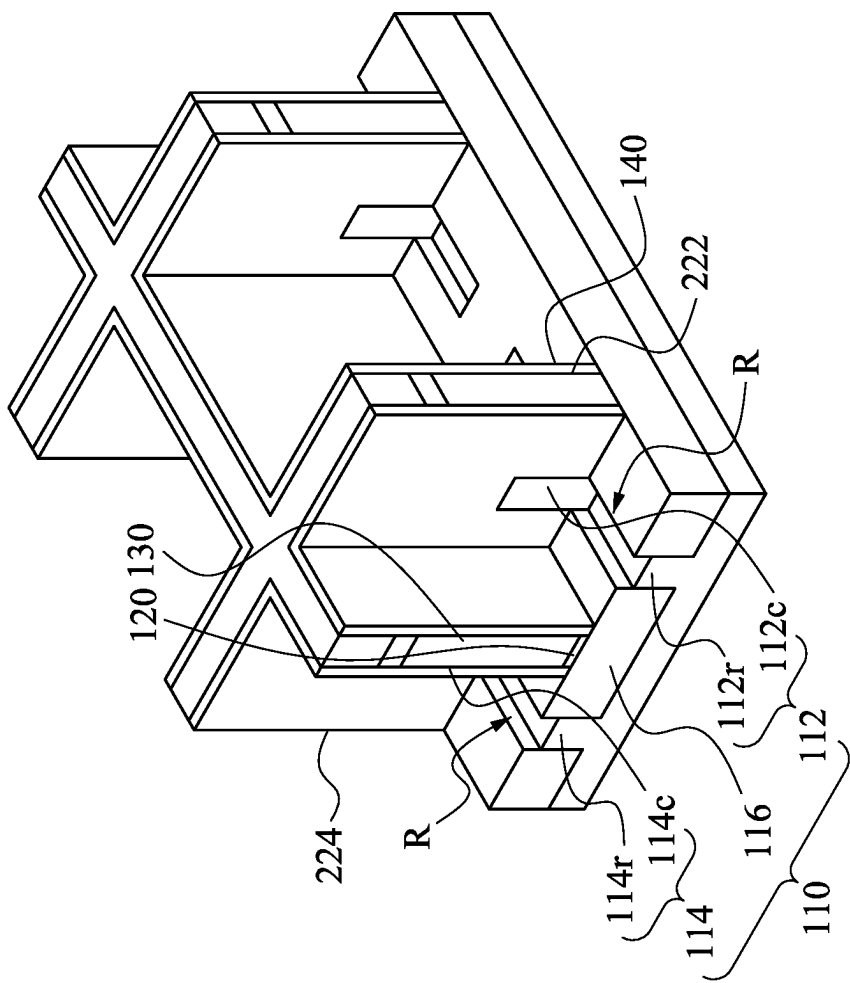


Fig. 8

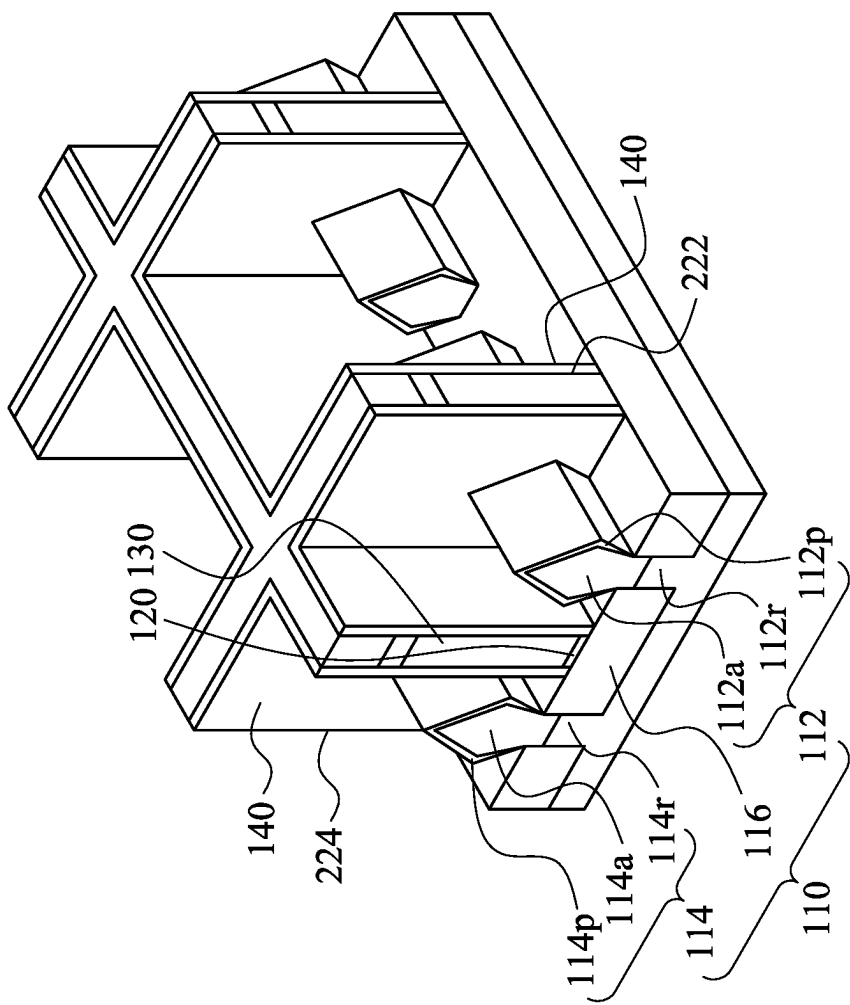


Fig. 9

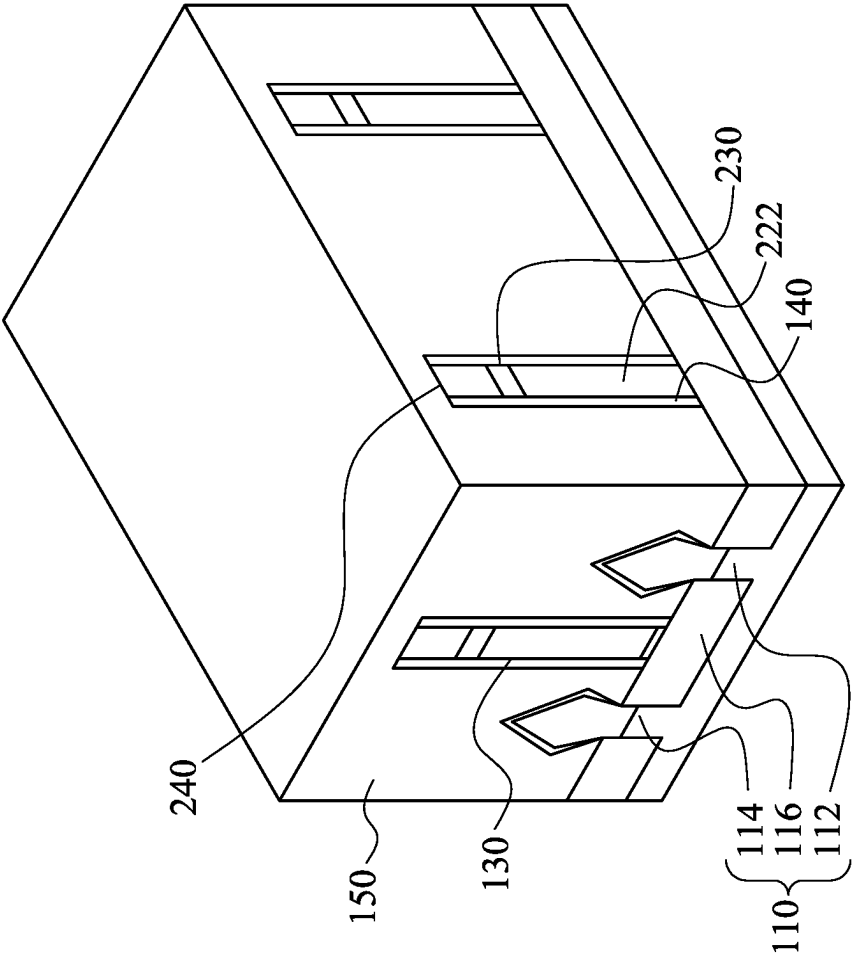


Fig. 10

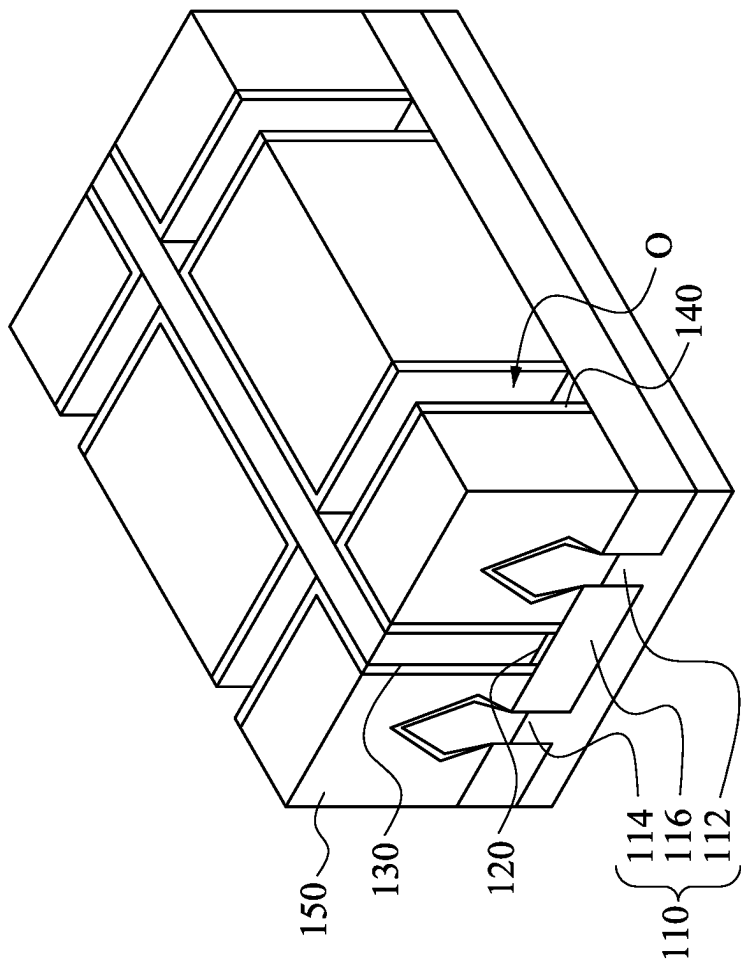


Fig. 11

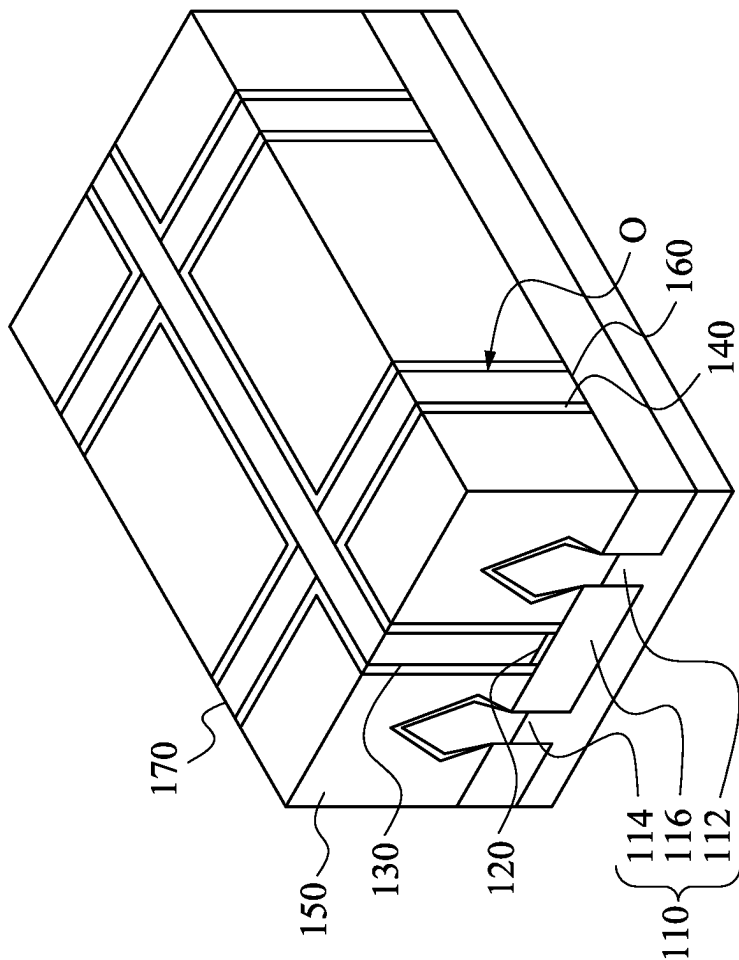


Fig. 12

1

SEMICONDUCTOR STRUCTURE AND MANUFACTURING METHOD THEREOF

PRIORITY CLAIM AND CROSS-REFERENCE

This application is a continuation of U.S. patent application Ser. No. 16/050,028, entitled “Semiconductor Structure and Manufacturing Method Thereof”, and filed on Jul. 31, 2018, which is a divisional of U.S. patent application Ser. No. 15/061,876 (now U.S. Pat. No. 10,510,608, issued Dec. 17, 2019), entitled “Semiconductor Structure and Manufacturing Method Thereof,” and filed on Mar. 4, 2016, which applications are incorporated herein by reference.

BACKGROUND

Double-gate metal-oxide-semiconductor field-effect transistors (MOSFETs) are MOSFETs that incorporate two gates into one device. These devices are also called fin-shaped field effect transistors (FinFETs) due to their structure including a thin “fin” extending from a substrate. Silicon based FinFETs can be fabricated using MOSFET technology. A FinFET is fabricated on a substrate with an overlying insulating layer with a thin “fin” extending from the substrate, for example, etched into a silicon layer of the substrate. The channel of the field effect transistor (FET) is formed in this vertical fin. A gate is provided over (e.g., wrapping) the fin. The double gate is beneficial in that there is a gate on both sides of the channel allowing gate control of the channel from both sides. Further advantages of FinFETs include reducing the short channel effect and higher current flow. Other FinFET architectures may include three or more effective gates.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. 1-12 are perspective views of a method for manufacturing a semiconductor structure at various stages in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

2

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

The singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising”, or “includes” and/or “including” or “has” and/or “having” when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof.

It will be understood that when an element is referred to as being “on” another element, it can be directly on the other element or intervening elements may be present therebetween. In contrast, when an element is referred to as being “directly on” another element, there are no intervening elements present. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by the person having ordinary skill in the art. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

FIGS. 1-12 are perspective views of a method for manufacturing a semiconductor structure **100** at various stages in accordance with some embodiments of the present disclosure.

Reference is made to FIG. 1. A substrate **110** is provided. The substrate **110** includes first fins **112** and second fins **114**. In some embodiments, the first fins **112** and the second fins **114** include silicon. It is noted that the numbers of the first fins **112** and the second fins **114** in FIG. 1 are illustrative and should not limit various embodiments of the present disclosure. A person having ordinary skill in the art may select suitable numbers for the first fins **112** and the second fins **114** according to actual situations.

In some embodiments, the substrate **110** may be made of a semiconductor material and may include, for example, a graded layer or a buried oxide therein. In some embodiments, the substrate **110** includes bulk silicon that may be undoped or doped (e.g., p-type, n-type, or combinations thereof). Other materials that are suitable for semiconductor device formation may be used. For example, germanium, quartz, sapphire, and glass could alternatively be used for the substrate **110**. Alternatively, the silicon substrate **110** may be an active layer of a semiconductor-on-insulator (SOI) substrate or a multi-layered structure such as a silicon-germanium layer formed on a bulk silicon layer.

The first fins **112** and the second fins **114** may be formed by, for example, patterning and etching the substrate **110** using photolithography techniques. In some embodiments, a layer of photoresist material (not shown) is formed over the

3

substrate **110**. The layer of photoresist material is irradiated (or exposed) in accordance with a pattern (the first fins **112** and the second fins **114**) and developed to remove a portion of the photoresist material. The remaining photoresist material protects the underlying material from subsequent processing steps, such as etching. Then, an etching process is performed to form the first fins **112** and the second fins **114**. It should be noted that other masks, such as an oxide or silicon nitride mask, may also be used in the etching process.

In some other embodiments, the first fins **112** and the second fins **114** may be epitaxially grown. For example, exposed portions of an underlying material, such as exposed portions of the substrate **110**, may be used in an epitaxial process to form the first fins **112** and the second fins **114**. A mask may be used to control the shape of the first fins **112** and the second fins **114** during the epitaxial growth process.

The substrate **110** further includes isolation structures **116**. The isolation structures **116**, which act as shallow trench isolation (STI) structures between the first fins **112** and the second fins **114**, may be formed by filling trenches between the first fins **112** and the second fins **114** with a flowable dielectric material, curing the flowable dielectric material, and then recessing the cured dielectric material. In yet some other embodiments, the isolation structures **116** are insulator layers of a SOI wafer.

In FIG. 2, an insulating layer **120** is formed over the first fins **112**, the second fins **114**, and the isolation structures **116**. In some embodiments, the insulating layer **120** may be made of silicon nitride. The insulating layer **120** may be formed using chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), other suitable processes, or combinations thereof. In addition, in some embodiments, the insulating layer **120** and the isolation structures **116** are made of different materials.

Then, a mask layer **210** is formed over the substrate **110** to cover the insulating layer **120**. In some embodiments, the mask layer **210** includes a semiconductor material, such as polysilicon or the like. The mask layer **210** may be deposited doped or undoped. For example, in some embodiments the mask layer **210** includes polysilicon deposited undoped by low-pressure chemical vapor deposition (LPCVD). The polysilicon may also be deposited by, for example, furnace deposition of an in-situ doped polysilicon. Alternatively, the mask layer **210** may include other suitable materials.

In FIG. 3, a gap **212** may be formed between the first fin **112** and the second fin **114** by, for example, patterning and etching the mask layer **210** using photolithography techniques. In some embodiments, a layer of photoresist material (not shown) is formed over the mask layer **210**. The layer of photoresist material is irradiated (or exposed) in accordance with a pattern (the gap **212**) and developed to remove a portion of the photoresist material. The remaining photoresist material protects the underlying material from subsequent processing steps, such as etching. Then, an etching process is performed to form the gap **212**.

Embodiments of this disclosure are not limited thereto. In some embodiments, a trilayer microlithographic process may be used here. Specifically, the mask layer **210** may be made of a $\text{SiC}_x\text{H}_y\text{O}_z$ material, and the mask layer **210** may be formed by spin coating. Before the gap **212** is formed between the first fin **112** and the second fin **114**, a layer of photoresist material (not shown) is formed over the mask layer **210**. Then, the layer of photoresist material is irradiated (or exposed) in accordance with a pattern (the gap **212**) and developed to remove a portion of the photoresist material. The remaining photoresist material protects the underlying material from subsequent processing steps, such as

4

etching. Then, the mask layer **210** may be etched to form the gap **212** by deep ultraviolet or by fluorine-base RIE (reactive-ion etching).

In some other embodiments, the mask layer **210** may be made of a $\text{C}_x\text{H}_y\text{O}_z$ material. Before the gap **212** is formed between the first fin **112** and the second fin **114**, a middle layer (not shown) is formed over the mask layer **210**, and then a layer of photoresist material (not shown) is formed over the middle layer, in which the middle layer may be made of a $\text{SiC}_x\text{H}_y\text{O}_z$ material and may be formed by spin coating. Then, the layer of photoresist material is irradiated (or exposed) in accordance with a pattern (the gap **212**) and developed to remove a portion of the photoresist material. The remaining photoresist material protects the underlying material from subsequent processing steps, such as etching. Then, the middle layer may be etched by deep ultraviolet or by fluorine-base RIE, and then the pattern of the middle layer may be transferred to the mask layer **210** to form the gap **212** by irradiation and development.

Briefly, the mask layer **210** may be made of a photoresist material in some embodiments, but the mask layer **210** and the layer of photoresist material used to define the gap **212** may be made of different materials.

Then, an insulating structure **130** is formed in the gap **212**. The insulating structure **130** is present on the insulating layer **120** and between the first fins **112** and the second fins **114**. The insulating structure **130** may include, for example, silicon dioxide (SiO_2), silicon nitride (SiN), silicon oxynitride (SiON), silicon carbide (SiC), aluminum oxide (Al_2O_3), copper oxide (Cu_xO_y , $x=1$ or 2 ; $y=1$, 2 , or 3), or M_xO_y ($\text{M}=\text{Ni}$, Zn , LaMn , or La_2Cu ; $x=1$ or 2 ; $y=1$, 2 , or 3). The insulating structure **130** may be formed using chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), other suitable processes, or combinations thereof. In addition, in some embodiments, the insulating layer **120** and the isolation structures **116** are made of different materials.

Reference is made to FIG. 3 and FIG. 4. The mask layer **210** is removed by etching, and the insulating layer **120** is also removed by etching while leaving a part of the insulating layer **120** between the isolation structure **116** and the insulating structure **130**. In other words, the insulating layer **120** is present between the isolation structure **116** and the insulating structure **130**. The insulating layer **120** between the isolation structure **116** and the insulating structure **130** can release stress and/or prevent electrons from tunneling. A ratio of a thickness of the insulating structure **130** and a thickness of the insulating layer **120** is substantially greater than about 10. Specifically, the thickness of the insulating structure **130** is in a range from about 10 nm to about 150 nm, and a width of the insulating structure **130** is in a range from about 15 nm to about 30 nm.

Specifically, when the mask layer **210** is made of a $\text{SiC}_x\text{H}_y\text{O}_z$ material, the mask layer **210** may be etched by deep ultraviolet or by fluorine-base RIE. When the mask layer is made of a $\text{C}_x\text{H}_y\text{O}_z$ material, the mask layer may be etched by irradiation and development.

In FIG. 5, a dummy gate layer **220** is formed on the substrate **110**. Specifically, the dummy gate layer **220** is deposited and then planarized, such that a height of insulating structure **130** and a height of the dummy gate layer **220** are substantially the same. The dummy gate layer **220** may include, for example, polysilicon, amorphous silicon, or the like. The deposition of the dummy gate layer **220** is performed by chemical vapor deposition (CVD), physical vapor deposition (PVD), other suitable processes, or com-

5

binations thereof. The planarization of the dummy gate layer **220** is performed by chemical mechanical polishing.

Then, a stop layer **230** is formed on the dummy gate layer **220** and the insulating structure **130**. The stop layer **230** may include, for example, silicon nitride, titanium nitride, or the like. The stop layer **230** may be formed using chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), other suitable processes, or combinations thereof.

Then, a hard mask layer **240** is formed on the stop layer **230**. The hard mask layer **240** may include, for example, silicon dioxide (SiO_2), silicon nitride (SiN), oxynitrides (SiON), silicon carbide (SiC), aluminum oxide (Al_2O_3), copper oxide (Cu_xO_y , $x=1$ or 2 ; $y=1, 2$, or 3), or M_xO_y ($\text{M}=\text{Ni}$, Zn , LaMn , or La_2Cu ; $x=1$ or 2 ; $y=1, 2$, or 3). The stop layer **230** may be formed using chemical vapor deposition (CVD), physical vapor deposition (PVD), other suitable processes, or combinations thereof.

Then, a bottom layer **250** is formed on the hard mask layer **240**. The bottom layer **250** may include, for example, a $\text{C}_x\text{H}_y\text{O}_z$ material. Then, as shown in FIG. 5 and FIG. 6, a trilayer microlithographic process may be used here to form first dummy gates **222** and second dummy gates **224**.

Specifically, a middle layer is deposited over the bottom layer **250**, and a layer of photoresist material (not shown) is formed over the middle layer, in which the middle layer may be made of a $\text{SiC}_x\text{H}_y\text{O}_z$ material and may be formed by spin coating. Then, the layer of photoresist material is irradiated (exposed) in accordance with a pattern (the first dummy gates **222** and the second dummy gates **224**) and developed to remove a portion of the photoresist material. The remaining photoresist material protects the underlying material from subsequent processing steps, such as etching. Then, the middle layer may be etched by deep ultraviolet or by fluorine-base RIE, and then the pattern of the middle layer is transferred to the bottom layer **250**. Subsequently, the dummy gate layer **220**, the stop layer **230**, and the hard mask layer **240** are patterned and etched to form the first dummy gates **222** and the second dummy gates **224** at opposite sides of the insulating structure **130**, in which the first dummy gates **222** and the second dummy gates **224** are respectively at least partially present on the first fins **112** and second fins **114**. After the first dummy gates **222** and the second dummy gates **224** are formed, the bottom layer **250** is removed by etching.

In some other embodiment, the bottom layer **250** may include, for example, polysilicon, amorphous silicon, or the like. The bottom layer **250** may be formed using chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), other suitable processes, or combinations thereof. Then, instead of using a trilayer microlithographic process, a simple microlithographic process may be used here. Specifically, a layer of photoresist material (not shown) is formed over the bottom layer **250**. The layer of photoresist material is irradiated (or exposed) in accordance with a pattern (the first dummy gates **222** and the second dummy gates **224**) and developed to remove a portion of the photoresist material. The remaining photoresist material protects the underlying material from subsequent processing steps, such as etching. Then, an etching process is performed to form the first dummy gates **222** and the second dummy gates **224**. After the first dummy gates **222** and the second dummy gates **224** are formed, the bottom layer **250** is removed by etching. In yet some other embodiments using a simple microlithographic process, the bottom layer **250** may not be formed.

6

Reference is made to FIG. 7. Spacers **140** are formed on side walls of the first dummy gates **222** and the second dummy gates **224** and side walls of the insulating structure **130**. In some embodiments, at least one of the spacers **140** include one or more layers, including silicon nitride, silicon oxynitride, silicon oxide, or other dielectric materials. The available formation methods include plasma enhanced chemical vapor deposition (PECVD), low-pressure chemical vapor deposition (LPCVD), sub-atmospheric chemical vapor deposition (SACVD), and other deposition methods.

Reference is made to FIG. 8. Portions of the first fins **112** and the second fins **114** exposed by the spacers **140**, the first dummy gates **222**, and the second dummy gates **224** are partially removed (or partially recessed) to form recesses **R** in the first fins **112** and the second fins **114**. In some embodiments, side walls of the recesses **R** are substantially vertically parallel to each other. In some other embodiments, the recesses **R** are formed with a non-vertical parallel profile.

At least one of the first fins **112** includes at least one channel portion **112c** and at least one recessed portion **112r**. The first dummy gates **222** and the spacers **140** cover the channel portions **112c**, and the recesses **R** are formed on the recessed portions **112r**. At least one of the second fins **114** includes at least one channel portion **114c** and at least one recessed portion **114r**. The second dummy gates **224** and the spacers **140** cover the channel portions **114c**, and the recesses **R** are formed on the recessed portion **114r**.

The recessing process may include a dry etching process, a wet etching process, and/or combinations thereof. The recessing process may also include a selective wet etching process or a selective dry etching process. A wet etching solution includes a tetramethylammonium hydroxide (TMAH), a $\text{HF}/\text{HNO}_3/\text{CH}_3\text{COOH}$ solution, or other suitable solution. The dry and wet etching processes have etching parameters that can be tuned, such as etchants used, etching temperature, etching solution concentration, etching pressure, source power, radio frequency (RF) bias voltage, RF bias power, etchant flow rate, and other suitable parameters. In some other embodiments, a wet etching solution may include NH_4OH , KOH (potassium hydroxide), HF (hydrofluoric acid), TMAH (tetramethylammonium hydroxide), other suitable wet etching solutions, or combinations thereof. In yet some other embodiments, a dry etching process may include a biased plasma etching process that uses a chlorine-based chemistry. Other dry etchant gasses include CF_4 , NF_3 , SF_6 , and He . Dry etching may also be performed anisotropically using such mechanisms as DRIE (deep reactive-ion etching).

Reference is made to FIG. 9. A plurality of epitaxy structures **112a** are respectively formed in the recesses **R** (see FIG. 8) and on the recessed portions **112r** of the first fins **112**, and a plurality of epitaxy structures **114a** are respectively formed in the recesses **R** and on the recessed portions **114r** of the second fins **114**. The epitaxy structures **112a** and **114a** protrude from the recesses **R**. In some embodiments, lattice constants of the epitaxy structures **112a** and **114a** are different from lattice constants of the channel portions **112c** and **114c** (see FIG. 8) of the first fins **112** and the second fins **114**, and thus the channel portions **112c** and **114c** are strained or stressed to enhance carrier mobility of the semiconductor device and the device performance. The epitaxy structures **112a** and **114a** may be formed using a cyclic deposition and etching (CDE) process. The CDE process includes an epitaxial deposition/partial etch process and repeating the epitaxial deposition/partial etch process at least once.

In some embodiments that the resulting metal-oxide-semiconductor (MOS) device is an nMOS device, the epitaxy structures **112a** and **114a** may be n-type epitaxy structures. In some embodiments that the resulting MOS device is an pMOS device, the epitaxy structures **112a** and **114a** may be p-type epitaxy structures. The n-type epitaxy structures may be made of SiP, SiC, SiPC, Si, III-V compound semiconductor materials or combinations thereof, and the p-type epitaxy structures may be made of SiGe, SiGeC, Ge, Si, III-V compound semiconductor materials, or combinations thereof. During the formation of the n-type epitaxy structures, n-type impurities such as phosphorous or arsenic may be doped with the proceeding of the epitaxy. For example, when the n-type epitaxy structures include SiC or Si, n-type impurities are doped. Moreover, during the formation of the p-type epitaxy structures, p-type impurities such as boron or BF₂ may be doped with the proceeding of the epitaxy. For example, when the p-type epitaxy structures includes SiGe, p-type impurities are doped. The epitaxy processes include CVD deposition techniques (e.g., vapor-phase epitaxy (VPE) and/or ultra-high vacuum CVD (UHV-CVD)), molecular beam epitaxy, and/or other suitable processes. The epitaxy process may use gaseous and/or liquid precursors, which interact with the composition of the first fins **112** and the second fins **114** (e.g., silicon). Thus, a strained channel can be achieved to increase carrier mobility and enhance device performance. The epitaxy structures **112a** and **114a** may be in-situ doped. If the epitaxy structures **112a** and **114a** are not in-situ doped, a second implantation process (i.e., a junction implant process) is performed to dope the epitaxy structures **112a** and **114a**. One or more annealing processes may be performed to activate the epitaxy structures **112a** and **114a**. The annealing processes include rapid thermal annealing (RTA) and/or laser annealing processes. In some embodiments, silicon caps **112p** and **114p** may be formed covering the epitaxy structures **112a** and **114a**. The silicon caps **112p** and **114p** may be made of undoped polysilicon or undoped amorphous silicon. The silicon caps **112p** and **114p** may be formed by ALD, CVD, plasma-enhanced chemical vapor deposition (PECVD), high density chemical vapor deposition (HDCVD), PVD, and the like.

Therefore, the first fins **112** with a new form, which include the channel portions **112c** (see FIG. 7), the recessed portions **112r**, and the epitaxy structures **112a**, are formed, and the second fins **114** with a new form, which include the channel portions **114c** (see FIG. 7), the recessed portions **114r**, and the epitaxy structures **114a**, are formed.

As shown in FIG. 10. An inter-layer dielectric (ILD) **150** is formed over the substrate **110** and covers the first dummy gate **222**, the second dummy gate **224**, and the insulating structure **130**. The ILD **150** may include a flowable dielectric material formed using, for example, flowable chemical vapor deposition (FCVD). The ILD **150** may also be a spin-on glass formed using spin-on coating. For example, the ILD **150** may include phospho-silicate glass (PSG), boro-silicate glass (BSG), boron-doped phospho-silicate glass (BPSG), tetraethyl orthosilicate (TEOS) oxide, TiN, SiOC, or other low-k non-porous dielectric materials.

Reference is made to FIG. 10 and FIG. 11, a planarization step is performed by, for example, chemical mechanical polishing (CMP). The CMP is performed to remove excess portions of the ILD **150**, the remaining hard mask layer **240**, the remaining stop layer **230**, and excess portions of the spacers **140**, in which excess portions of the ILD **150** and the spacers **140** are over the top surfaces of the first dummy gates **222**, the second dummy gates **224**, and the insulating

structure **130**. In other words, the CMP stops on the top surfaces of the first dummy gates **222**, the second dummy gates **224**, and the insulating structure **130**.

Then, the first dummy gates **222** and the second dummy gates **224** are removed.

Openings **O** are formed as a result of the removal of the first dummy gates **222** and the second dummy gates **224**. After the removal of the first dummy gates **222** and the second dummy gates **224**, parts of side walls of the insulating structure **130** are exposed by the spacers **140**.

Reference is made to FIG. 11 and FIG. 12. First gates **160** and second gates **170** are formed in the openings **O**, such that the first gates **160** are present on the first fins **112**, and the second gates **170** are present on the second fins **114**. In other words, the first dummy gates **222** (see FIG. 10) and the second dummy gates **224** (see FIG. 10) are replaced respectively with the first gates **160** and the second gates **170**. Then, the first gates **160** and the second gates **170** are disposed at opposite sides of the insulating structure **130**, and the spacers **140** are present on side walls of the first gates **160** and the second gates **170**. In addition, the spacers **140** are substantially absent between the insulating structure **130** and the first gates **160** and between the insulating structure **130** and the second gates **170**. In other words, the end surfaces of the first gates **160** and the second gates **170** facing the insulating structure **130** are uncovered by the spacers **140**, and the insulating structure **130** is in contact with the first gates **160** and the second gates **170**.

Specifically, in some embodiments, a gate dielectric layer, a diffusion barrier layer, a metal layer, a block layer, a wetting layer, and a filling metal are sequentially formed in the openings **O**. Then, a planarization step (for example, chemical mechanical polish (CMP)) is performed to remove excess portions of the gate dielectric layer, the diffusion barrier layer, the metal layer, the block layer, the wetting layer, and the filling metal, in which the excess portions are over the ILD **150**. Therefore, the remaining portions of the gate dielectric layer, the diffusion barrier layer, the metal layer, the block layer, the wetting layer, and the filling metal form the first gates **160** and the second gates **170**. In other words, at least one of the first gates **160** and the second gates **170** may include the gate dielectric layer, the diffusion barrier layer, the metal layer, the block layer, the wetting layer, and the filling metal.

In some embodiments, the gate dielectric layer includes an interfacial layer (IL, the lower part of the gate dielectric layer), which is a dielectric layer. In some embodiments, the IL includes an oxide layer, such as a silicon oxide layer, which may be formed through a thermal oxidation of the substrate **110**, a chemical oxidation, or a deposition step. The gate dielectric layer may also include a high-k dielectric layer (the upper part of the gate dielectric layer) including a high-k dielectric material, such as hafnium oxide, lanthanum oxide, aluminum oxide, or combinations thereof. The dielectric constant (k-value) of the high-k dielectric material is higher than about 3.9, and may be higher than about 7, and sometimes as high as about 21 or higher. The high-k dielectric layer is overlying, and may contact, the IL.

In some embodiments, the diffusion barrier layer includes TiN, TaN, or combinations thereof. For example, the diffusion barrier layer may include a TiN layer (the lower part of the diffusion barrier layer), and a TaN layer (the upper part of the diffusion barrier layer) over the TiN layer.

When the first gates **160** or the second gates **170** form n-type metal-oxide-semiconductor (MOS) devices, the metal layer is in contact with the diffusion barrier layer. For example, in the embodiments in which the diffusion barrier

layer includes a TiN layer and a TaN layer, the metal layer may be in physical contact with the TaN layer. In alternative embodiments in which the first gates **160** or the second gates **170** form p-type metal-oxide-semiconductor (MOS) devices, an additional TiN layer is formed between, and in contact with, the TaN layer (in the diffusion barrier layer) and the overlying metal layer. The additional TiN layer provides the work function suitable for the pMOS device, which work function is higher than the mid-gap work function (about 4.5 eV) that is in the middle of the valance band and the conduction band of silicon. The work function higher than the mid-gap work function is referred to as a p-work function, and the respective metal having the p-work function is referred to as a p-metal.

The metal layer provides the work function suitable for the nMOS device, which work function is lower than the mid-gap work function. The work function lower than the mid-gap work function is referred to as an n-work function, and the respective metal having the n-work function may be referred to as an n-metal. In some embodiments, the metal layer is an n-metal having a work function lower than about 4.3 eV. The work function of the metal layer may also be in a range from about 3.8 eV to about 4.6 eV. The metal layer may include titanium aluminum (TiAl) (which may include, or free from or substantially free from other elements) in accordance with some embodiments. The formation of the metal layer may be achieved through physical vapor deposition (PVD). In accordance with some embodiments of the present disclosure, the metal layer is formed at room temperature (for example, from about 20° C. to about 25° C.). In alternative embodiments, the metal layer is formed at an elevated temperature higher than the room temperature, for example, higher than about 200° C.

The block layer may include TiN in some embodiments. The block layer may be formed using atomic layer deposition (ALD).

The wetting layer has an ability to adhere (and wet) the subsequently formed filling metal during the reflow of the filling metal. In some embodiments, the wetting layer is a cobalt layer, which may be formed using atomic layer deposition (ALD) or chemical vapor deposition (CVD).

The filling metal may include aluminum, an aluminum alloy (e.g., titanium aluminum), tungsten, or copper, which may also be formed using physical vapor deposition (PVD), chemical vapor deposition (CVD), or the like. The filling metal may be reflowed to fully fill the remaining openings **O**. The formation of the wetting layer improves the wetting of the filling metal to the underlying layers.

In some embodiments, when the mask layer **210** and the dummy gate layer **220** are made of the same material, for example, polysilicon, the mask layer **210** may not be removed after the insulating structure **130** is formed. Instead, the mask layer **210** is directly used as the dummy gate layer **220** of FIG. 5. Since the mask layer **210** is not removed, the insulating layer **120** is not removed as well after the insulating structure **130** is formed. However, the insulating layer **120** is removed when the dummy gate layer **220** is patterned as shown in FIG. 6 and when the first dummy gates **222** and the second dummy gates **224** are removed as shown in FIG. 11, while leaving the part of the insulating layer **120** between the isolation structure **116** and the insulating structure **130**. Therefore, the resulting structure of the semiconductor structure **100** is substantially the same as that of the semiconductor structure **100** in the aforementioned embodiments.

In some embodiments of the present disclosure, the insulating structure **130** is formed before the first gates **160** and

the second gates **170** are formed. Therefore, even if a gap between the first gates **160** and the second gates **170** is narrow, the insulating structure **130** can still be formed to isolate the first gates **160** and the second gates **170**. Furthermore, during formation of the first gates **160** and the second gates **170**, the insulating structure **130** has been formed, and thus the first gates **160** and the second gates **170** can be formed without concern about bridging the first gates **160** and the second gates **170**.

According to some embodiments of the present disclosure, a semiconductor structure includes a first fin, a second fin, a first gate, a second gate, at least one spacer, and an insulating structure. The first gate is present on the first fin. The second gate is present on the second fin. The spacer is present on at least one side wall of at least one of the first gate and the second gate. The insulating structure is present between the first fin and the second fin, in which the spacer is substantially absent between the insulating structure and said at least one of the first gate and the second gate.

According to some embodiments of the present disclosure, a semiconductor structure includes at least one fin, at least one gate, at least one isolation structure, an insulating structure, and at least one insulating layer. The gate is present on the fin. The isolation structure is present adjacent to the fin. The insulating structure is present on the isolation structure. The insulating layer is present between the insulating structure and the isolation structure, in which the insulating layer and the insulating structure are made of different materials.

According to some embodiments of the present disclosure, a method for manufacturing a semiconductor structure includes the following steps. A mask layer is formed to cover a first fin and a second fin. The mask layer is patterned to form a gap between the first fin and the second fin. An insulating structure is formed in the gap. First and second gates are formed at opposite sides of the insulating structure, in which the first and second gates are respectively at least partially present on the first and second fins.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A method comprising:

- forming a first fin and a second fin extending from a substrate;
- depositing an insulating layer over the first fin and the second fin;
- depositing a mask layer over the insulating layer;
- etching the mask layer between the first fin and the second fin;
- forming an insulating structure in the mask layer over the insulating layer and between the first fin and the second fin;
- removing the mask layer;
- forming a first dummy gate structure over the first fin after forming the insulating structure;
- removing the first dummy gate structure; and

11

after removing the first dummy gate structure, forming a first gate over the first fin.

2. The method of claim 1, further comprising forming a spacer along a sidewall of the first dummy gate structure, a sidewall of the insulating structure, and a sidewall of the insulating layer.

3. The method of claim 1, wherein the first dummy gate structure is formed along a sidewall of the insulating layer and a sidewall of the insulating structure.

4. The method of claim 1, further comprising forming a shallow trench isolation region on the substrate and surrounding the first fin and the second fin, wherein the insulating layer is deposited on the shallow trench isolation region.

5. The method of claim 1, wherein the insulating structure comprises a metal oxide.

6. The method of claim 1, wherein the insulating layer comprises silicon nitride.

7. A method, comprising:

depositing an insulating layer on a first fin and a second fin;

depositing a mask layer on the insulating layer;

etching the mask layer to form an opening exposing the insulating layer between the first fin and the second fin;

filling the opening with an insulating structure, wherein the insulating structure is on a first portion of the insulating layer, and wherein the mask layer is on a second portion of the insulating layer;

removing the mask layer and the second portion of the insulating layer;

forming a sacrificial structure on the first fin;

forming a dielectric layer on a sidewall of the sacrificial structure, a sidewall of the insulating structure, and a sidewall of the insulating layer;

removing the sacrificial structure; and

forming a gate structure over the first fin and adjacent to the dielectric layer.

8. The method of claim 7, further comprising etching the first fin using the dielectric layer and the sacrificial structure as masks.

9. The method of claim 8, further comprising forming a first epitaxy structure over the first fin, wherein the first epitaxy structure contacts the dielectric layer.

10. The method of claim 9, further comprising forming a first silicon cap layer over the first epitaxy structure, wherein the first silicon cap layer contacts the dielectric layer.

11. The method of claim 7, further comprising forming an interlayer dielectric on the first fin, the second fin, the dielectric layer, and the gate structure, wherein the dielectric layer is between the interlayer dielectric and each of the gate structure, the insulating layer, and the insulating structure.

12. The method of claim 11, further comprising planarizing the interlayer dielectric, the dielectric layer, the insulating structure, and the gate structure.

13. The method of claim 7, further comprising forming an interlayer dielectric on the first fin, the second fin, the

12

dielectric layer, and the sacrificial structure, wherein the dielectric layer is between the interlayer dielectric and each of the sacrificial structure, the insulating layer, and the insulating structure, and wherein the sacrificial structure is removed after forming the interlayer dielectric.

14. A method, comprising:

depositing a dielectric layer on a first semiconductor fin and a second semiconductor fin;

forming a mask layer over the dielectric layer;

etching the mask layer to form a gap between the first semiconductor fin and the second semiconductor fin, wherein the gap exposes the dielectric layer;

forming a dielectric fin filling the gap in the mask layer; after forming the dielectric fin, forming a first dummy gate structure over the first semiconductor fin and

forming a second dummy gate structure over the second semiconductor fin;

after forming the dielectric fin, forming a spacer in contact with a side surface of the dielectric layer, and in contact with a side surface of the dielectric fin, wherein the spacer extends along the side surface of the dielectric fin from the first dummy gate structure to the second dummy gate structure;

forming a first epitaxy structure on the first semiconductor fin and a second epitaxy structure on the second semiconductor fin; and

after forming the first epitaxy structure and the second epitaxy structure, replacing the first dummy gate structure and the second dummy gate structure with a first metal gate and a second metal gate.

15. The method of claim 14, wherein the mask layer is formed on a first portion of the dielectric layer, wherein the dielectric fin is formed on a second portion of the dielectric layer.

16. The method of claim 15, further comprising removing the mask layer and the first portion of the dielectric layer, wherein the second portion of the dielectric layer remains after removing the mask layer and the first portion of the dielectric layer.

17. The method of claim 14, wherein the mask layer comprises polysilicon, wherein the dielectric layer comprises silicon nitride, and wherein the dielectric fin comprises an oxide.

18. The method of claim 14, further comprising forming a shallow trench isolation region between the first semiconductor fin and the second semiconductor fin, wherein the dielectric layer is deposited on the shallow trench isolation region.

19. The method of claim 14, further comprising:

etching the first semiconductor fin to form a first recess; epitaxially depositing the first epitaxy structure in the first recess; and

forming a first silicon cap over the first epitaxy structure.

20. The method of claim 14, wherein the spacer is not present between the dielectric fin and the first metal gate.

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